• Step 2: Move crankshaft rearward to limits of travel.
• Step 3: Then, move crankshaft forward to limits of travel.
• Step 4: Wedge an appropriate tool between the rear of the cylinder block **(NOT BED PLATE)** and the rear crankshaft counterweight. This will hold the crankshaft in it's furthest forward position.
• Step 5: Install and tighten bolts (1–10) in sequence shown in (Fig. 53) to 41 N·m (30 ft. lbs.).
• Step 6: Remove wedge tool used to hold crankshaft.

(10) Tighten bolts (1–10) again to 41 N·m (30 ft. lbs.) **PLUS** 1/4 turn in sequence shown in (Fig. 53).

(11) Install main bearing bedplate to engine block bolts (11–20), and torque each bolt to 28 N·m (20 ft. lbs.) in sequence shown in (Fig. 53).

(12) After the main bearing bedplate is installed, check the crankshaft turning torque. The turning torque should not exceed 5.6 N·m (50 in. lbs.).

CRANKSHAFT OIL SEAL - FRONT

REMOVAL

(1) Remove the crankshaft vibration damper (Fig. 54). (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL)

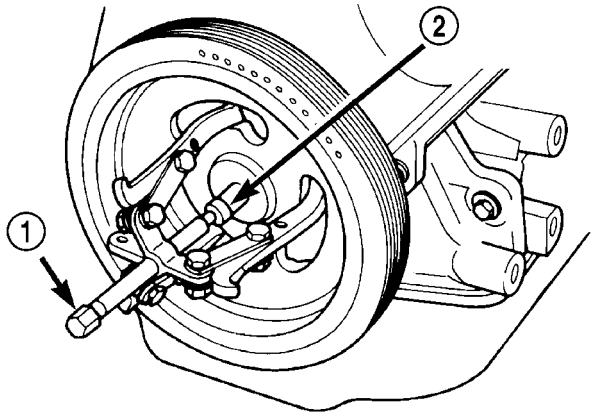

Fig. 54 Crankshaft Vibration Damper - Removal - Typical

1 - SPECIAL TOOL 1026 3-JAW PULLER
2 - SPECIAL TOOL 6827–A INSERT

(2) Remove timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(3) Remove crankshaft sprocket using Special Tool 6793 and insert C-4685-C2 (Fig. 55).

CAUTION: Do not nick shaft seal surface or seal bore.

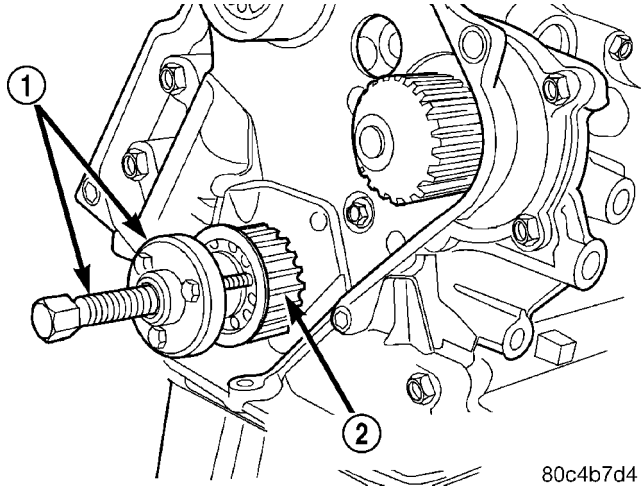

Fig. 55 Crankshaft Sprocket - Removal

1 - SPECIAL TOOL 6793
2 - CRANKSHAFT SPROCKET

(4) Using Tool 6771 to remove front crankshaft oil seal (Fig. 56). Be careful not to damage the seal surface of cover.

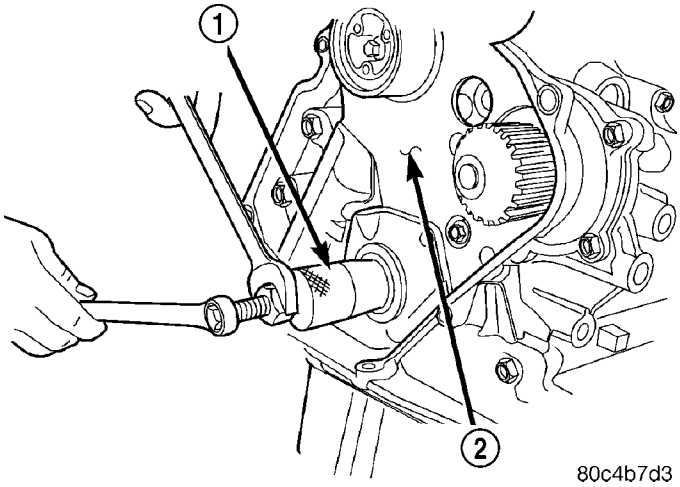

Fig. 56 Front Crankshaft Oil Seal - Removal

1 - SPECIAL TOOL 6771
2 - REAR TIMING BELT COVER

INSTALLATION

(1) Install new seal by using Special Tool 6780 (Fig. 57).

(2) Place seal into opening with seal spring towards the inside of engine. Install seal until flush with cover.